

Intercircuit: Electroplating with Cavities for Fast Fabrication of Complex and High-Performance 3D Circuits

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ABSTRACT

In Human-Computer Interaction, multi-material 3D printing is increasingly recognized for its capacity to produce conductive interactive objects. While various fabrication techniques have emerged in this domain, achieving the integration of highly conductive 3D structures within printed objects continues to pose a significant technical hurdle. Our study introduces Intercircuit, a novel integrated technique developed explicitly for fabricating highly conductive interactive objects. This method employs multi-material printing to form unified components along with their embedded structures of lower conductivity, enhancing conductivity through targeted plating. Distinct from traditional multi-material printing and surface conductivity augmentation methods, the Intercircuit method facilitates the creation of complex 3D circuitry while ensuring superior conductivity. Furthermore, this study introduces a supportive design tool aimed at aiding users in crafting conductive frameworks. Moreover, the practical application of this method is further elucidated through a series of case studies.

CCS CONCEPTS

• **Human-centered computing** → **Interaction devices.**

KEYWORDS

3D printed electronics; Selective Electroplating; conductive filament;

ACM Reference Format:

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1 INTRODUCTION

In the Human-Computer Interaction field, integrating functional circuits into 3D-printed objects has garnered significant interest among researchers. Additive manufacturing techniques are pivotal for rapid prototyping, offering benefits such as reduced design and manufacturing stages, accelerated product development, and increased adaptability to market demands.

Numerous methods exist for integrating circuits into interactive objects, broadly categorized into three groups. The first involves using conductive PLA in multi-material printing for complex three-dimensional circuitry. However, this is hampered by the high electrical resistance of current conductive materials and the interlayer resistance introduced during printing, limiting these methods' applications primarily to sensing and display purposes. The second category encompasses methods [15] like extruding low-melting-point metals [6] or embedding copper wires ultrasonically [5], enabling high-performance circuit printing [7]. Yet, these require costly, complex, and specialized equipment, often inaccessible to HCI researchers and the creative community. An alternative technique is adding conductive materials to the surface of 3D printed

objects, such as affixing conductive foils, transferring circuits [12], or applying conductive inks. However, these are predominantly surface-level applications, unsuitable for complex 3D circuit structures, and prone to damage, external interference, and potential safety hazards. Moreover, they often necessitate meticulous manual work, impacting stability and repeatability. Our work introduced Intercircuit, a method for integrally molding interactive objects with highly conductive internal 3D circuits. Unlike existing approaches, our method utilizes accessible printing materials and readily available resources to facilitate the construction of intricate internal conductive structures, thus broadening the scope of integrally fabricated interactive objects. Our method enables HCI researchers to rapidly prototype interactive objects with complex internal 3D circuits using consumer-grade multicolor 3D printers. We achieve optimal conductivity by constructing internal conductive pathways and cavities, subsequently enhanced through copper plating in a solution. Additionally, we have developed an assistive design tool to guide users in designing these conductive pathways and cavities and incorporating necessary accessories.

The main contributions of our paper are:

- Providing a more accessible manufacturing process for highly conductive interactive objects and validating the feasibility of various conductive structures.
- Introducing a circuit design tool to assist users in designing 3D models.
- Showcasing applications developed through our method, encouraging users to explore broader application scenarios.

2 RELATED WORK

2.1 Multi-material Printing for Manufacturing Interactive Objects

Poly(lactic acid)(PLA), as the most commonly used printing consumable, is the cornerstone of FDM printing technology. The advent of conductive PLA has enabled the creation of three-dimensional circuit structures in interactive objects. However, current commercial conductive PLA filaments exhibit high resistance, limiting their use mostly to capacitive sensors (such as Capricate [16], CAPath [8], Oh, Snap! [17], PrintPut [18]). High-performance materials like multi3D's Electrifi [13], with a resistivity of $0.006\Omega/\text{cm}$, are available but costly, priced at approximately 1700 USD per kilogram. These materials support only limited print speeds (15-45mm/s) and require printing at 150°C to maintain optimal conductivity. Despite good in-plane conductivity, they present high interlayer resistance, complicating multi-material printing and supporting complex 3D structures in tandem with standard PLA. Consequently, our approach emphasizes enhancing common high-resistance conductive PLA through copper plating, an affordable and accessible option for most researchers. It supports a broader range of interactive objects due to its strong printability.

2.2 Fabrication of High-Performance Circuits

To enhance the conductivity of interactive objects, HCI researchers have experimented with combining metal wires, foils, or highly conductive materials like conductive inks and silver pastes with 3D-printed objects via surface bonding or internal embedding. For instance, Optomec employs e-ink deposition on object surfaces, and

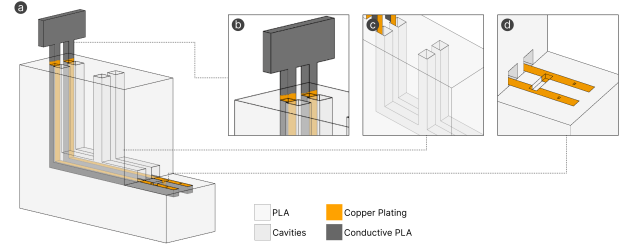


Figure 1: a) Integration examples of Intercircuit; b)Clamping piece for plating; c) Internal cavities and ventilation hole; d) Electronic component slots

Adams et al. [1] have printed small antennas on hemispherical glass substrates. These methods necessitate specialized multi-axis extrusion equipment and complex algorithms, while surface-exposed circuits are vulnerable to environmental impacts and human interaction hazards. Jose et al. [5] embedded copper wires in FDM prints using ultrasound, and Voxel8, now in commercial production, integrates conductive silver ink extrusion with FDM printing. However, these techniques depend on expensive, bespoke equipment, presenting significant entry barriers. Techniques like Foldio [14] and PrintGami [3] involve creating interactive objects by folding printed circuits into 3D models. Still, the fragility of paper material compromises the durability of the end product, limiting its application scope. Selective copper plating of conductive filaments has emerged as a promising avenue, as evidenced by studies demonstrating conductivity enhancement through copper plating [9–11]. Freddie Hong et al. [4] attempted to create 3D objects with selectively plated planar circuits using a heat bending process. Yet, these circuits remained surface-based, supporting only up to two wiring layers.

This paper investigated a method to construct high-performance internal circuits using common conductive PLA. Unlike previous approaches, we utilize easily accessible materials and standard printing equipment while enabling the construction of complex three-dimensional circuits. The circuit's performance is significantly improved through copper plating, facilitating its use in high-power applications and broadening the potential uses of conductive PLA.

3 BASIC STRUCTURE

Intercircuit is a method for constructing internal circuits utilizing multi-material printing. Its fundamental structure is depicted in Figure 1, encompassing an internal conductive track, an internal cavity and ventilation hole (Figure 1c), a clamping piece (Figure 1b), and a slot (Figure 1d). We employ conductive PLA infused with 15% carbon nanotubes to construct trajectories with low conductivity, featuring a resistivity of $120\Omega/\text{cm}$. Additionally, we build cavity structures adjacent to the conductive paths, facilitating the entry of plating solutions into the object's interior for effective plating. Vertical ventilation holes are added where bubble buildup may occur to prevent localized plating failures caused by gas buildup. Finally, the clamping pieces at the end of the conductive track allow the conductive portion of the conductive part to have a larger contact area

with the powered crocodile clips and keep the energized position above the liquid level.

In the electroplating process, the conductive PLA material is used as the cathode, while the copper foil acts as the anode, and both electrodes are submerged in a copper plating solution. A direct current is applied to both electrodes and the current flows from the anode to the cathode, causing copper metal ions to be transferred from the solution to the cathode of the conductive material in the printed object, resulting in the deposition of copper on the surface of the conductive material. Copper plating on conductive PLA surfaces can significantly improve the conductivity of an object by several orders of magnitude.

In this way, a high-performance three-dimensional structured circuit with low resistance can be obtained inside the object, and the details of how to design and fabricate such a structure will be described in Sections 4 and 5.

4 FABRICATION PIPELINE

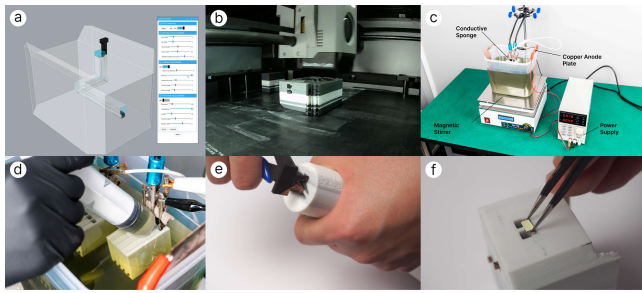


Figure 2: Manufacturing workflow: a) Model design using parametric assisted design tools; b) Printing of the model using PLA and conductive PLA; c) Plating in plating solution; d) Rinsing of the cavities; e) Remove the clamping piece; f) Placement of electrical components

In this subsection, we describe how internal circuit and cavity design can be performed with the help of auxiliary design tools and how high-resistance conductive traces from multimaterial prints can be transformed into low-resistance circuits by selective plating.

1. Model design, the user will first import the basic model into the rhino environment. The user needs to manually establish the spatial polyline segments as the internal conductive structure of the guideline; our auxiliary design tools allow users to select these spatial polylines to generate printable spatial conductive trajectories and cavity structures (Figure 2a). Additionally, users can use the tools to add features such as clamping tabs and vent holes.

2. Multi-material 3D Printing: Subsequently, users can proceed with multi-material printing (Figure 2b) using a standard consumer-grade multi-color printer. For our project, we utilized the Bambu Lab X1 FDM 3D printer with a 0.4mm printhead [2]. The slicing setup used 100% internal fill and linear fill modes. The number of filaments flushes was chosen to be 600mm to prevent potential problems with reduced conductivity caused by filaments staining each other. Regarding material selection, we employed Bambu Lab's official white PLA basic filaments and a commercially available

conductive PLA (from Hangzhou ZhuoXinPu materials), which is affordable and nearly equivalent in cost to regular PLA. This conductive PLA is readily accessible and shares similar printing parameters with standard PLA, allowing for high-speed printing of 200mm/s.

3. Copper Plating. Our project's plating solution consists of copper sulfate, sodium hydroxide, complexing agents, brighteners, and stabilizers. Among them, $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ (160 g/L) provides the required copper ions for electroplating, NaOH (20 g/L) regulates the alkalinity of the electroplating solution and provides the necessary alkaline environment, EDTA (12 g/L) mainly acts as a complexing agent and plays a certain role as an accelerator and stabilizer, and sodium p-benzenesulfonate $\text{NaC}_6\text{H}_4\text{SO}_3$ (25 mg/L) acts as a brightener.

In the preparation, an appropriate amount of purified water is first put into the reaction tank, and $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ is added and stirred evenly. Then add NaOH slowly, continuing to stir well. Next, add EDTA and continue stirring until completely dissolved. Finally, add $\text{NaC}_6\text{H}_4\text{SO}_3$ and stir well. Special attention must be paid to safety during operation to prevent corrosion and irritation of the skin by alkaline liquids.

We recommend flushing the cavities with a syringe to avoid the buildup of air bubbles in the internal cavities (Figure 2d). Based on our experience with copper plating, we began with a constant current of 0.04 A at a room temperature of 25°C, continuing until a visible thin copper layer formed over the entire circuit length (about 1 hour). The current was then reduced to 0.02 A (about 4 hours). A magnetic stirrer was used at 400 rpm throughout the plating process to ensure even distribution. The power supply's positive terminal was connected to a copper sheet, and the negative was attached to the component being plated. When connecting components with crocodile clips, we covered the conductive sponge in the holding part to enhance the electrical connection (Figure 2c).

4. Post-processing and Component Assembly: After plating, users need to remove the clamping piece (Figure 2e) and complete the assembly of components either by plugging or soldering (Figure 2f).

5 DESIGN INTERFACE

Our objective is to aid users in designing models with internal three-dimensional conductive structures. Modeling such complex structures typically involves intricate stretching and boolean operations. To streamline this process, we designed a plugin that automates the modeling of lines and cavities, thereby reducing repetitive tasks. The plugin was developed on the Rhino and Grasshopper platforms, with various functions and adjustable parameters:

1. Generate Conductive Trajectories (Figure 3a). First, users can build the basic model in Rhino's design window and then draw the spatial polyline needed for the conductive trajectory. By selecting the model and polyline in the panel using "object" and "line," users can generate the corresponding conductive trajectory. The trajectory's cross-sectional size is adjustable.

2. Create Cavities Around Trajectories. With the conductive trajectory generation, the plugin can simultaneously create cavities close to the trajectory. Users can opt to generate cavities in any

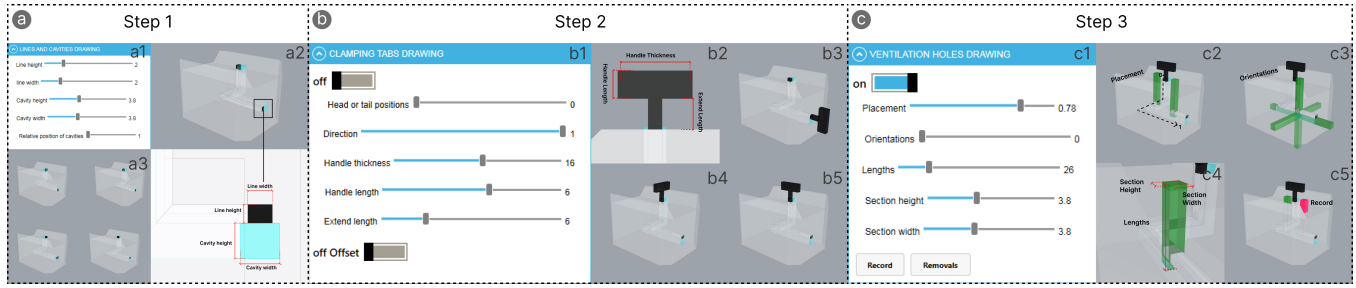


Figure 3: Main functions and design steps of the plug-in: a) Generation and adjustment of the internal conductive structure as well as the internal cavities; b) Generation and adjustment of the clamping structure; c) Generation and adjustment of the ventilation channel

of the four directions surrounding the conductive trajectory. The cavity's cross-sectional size is also adjustable (Figure 3a).

3. Generate Clamping Structure (Figure 3b). After the previous two steps, users can create clamping structures at either end of the conductive trajectory. The length and size of this clamping structure are adjustable to meet practical requirements.

4. Create Ventilation Holes. Ventilation structures can be generated along the cavity track in one direction, according to the user's choice. These structures can be individually produced, with the "Placement" slider to adjust position, "Record" to record, and "Removals" to remove. The ventilation channel's length is also modifiable to suit actual needs.

5. Conduct Boolean Operations. The plugin can automatically perform boolean operations on the model. First, it can add a boolean between conductive trajectories and clamping pieces. Then, it can create a boolean subtraction combined with the target model, the cavity, and the ventilation holes. Finally, the user performs a manual Boolean subtraction of the model at the location where the component needs to be placed to generate the slot for the component.

6. Export the Model. Once the target model is finalized, click "Export" to get the final file of the boolean model containing the conductive trajectories and clamping pieces.

6 INTERNAL CONDUCTIVE STRUCTURE TESTING

We conducted tests on various internal circuit structures to validate our method's capabilities and offer detailed guidance for other designers. These tests led to the creation of a comprehensive table library, described as follows:

6.1 Cross-sectional Area of Conductive Paths and Internal Cavities

The cross-sectional area of the conductive track determines the conductivity of the conductive path before plating, while the cross-sectional area of the cavity determines the ease of plating solution entry and mixing exchange, which together determine the ease of plating and plating quality. Balancing resolution and conductivity, we chose a conductive track with a cross-section of 3mm*3mm. A test piece (Figure 4a) with varying cavity cross-sectional areas was designed, featuring square cross-sections. These squares ranged from 3mm to 6mm sides, increasing by 0.2mm from left to right. Our

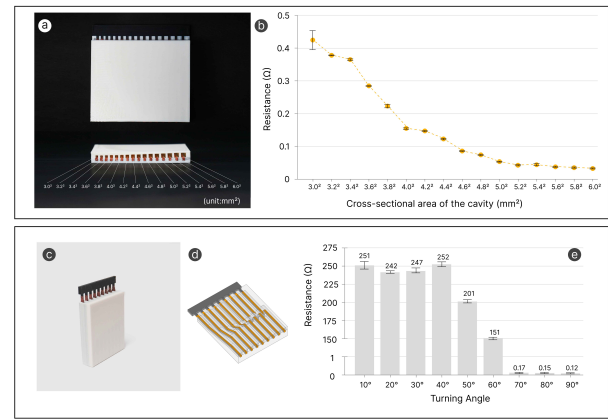


Figure 4: a) Cavity cross-sectional area test pieces; b) Resistance test results for different cavity cross-sectional areas; c) Internal turning angle test pieces; d) X-ray sketch of the internal turning angle test piece; e) Resistance test results for different internal turning angles

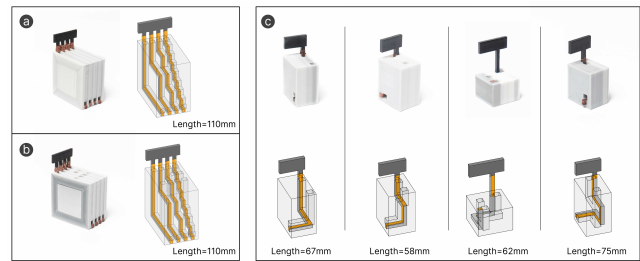


Figure 5: a) Bending complexity test piece without ventilation holes; b) Bending complexity test piece with ventilation holes; c) Space bending complexity test pieces

tests showed that for a 100mm length internal plating trajectory, resistance tends to stabilize when the cavity's cross-sectional area exceeds 4.8mm² (Figure 4b).

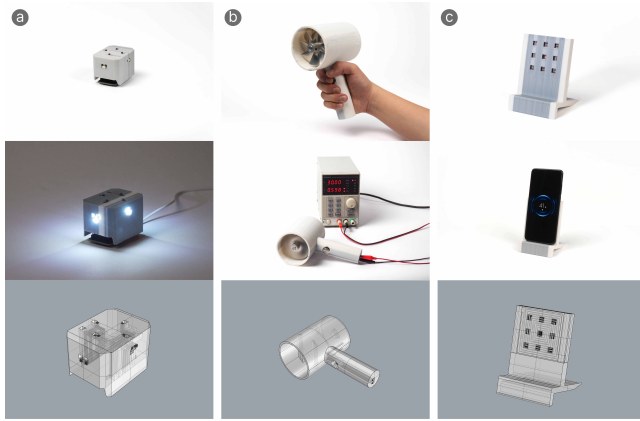


Figure 6: Cases: a) Multi-directional lamp; b) Handheld blower; c) Wireless charging phone holder

6.2 Conductive Track Turning Angle

For the circuit's turning angles (Figure 4c), after testing ten sets of plated parts that had been plated for 5 hours (the end-to-end distance of the conductive track was 100mm), we discovered that lines close to 90° offer better conductivity and occupy less space. They are also more conducive to drawing and layout. Therefore, we recommend designing turns at 90° .

6.3 Bending Complexity in the Same Plane

We verified the bending complexity of the method. For the experimental blocks with edge length of 55mm and conductive track length of 110mm, we set different numbers of bends to verify whether it will fail due to the number of turns, the number of bends is 1, 3, 7, and 15 times. (Figure 5a), When the experiment was carried out, two groups of test pieces with ventilation holes (Figure 5b) and without ventilation holes were used, with a total of six test pieces, divided into three different plating lengths (3h, 4h, 6h) on the conductivity of the line for testing. The results show that the test pieces with ventilation holes completed conduction (resistance below 0.3Ω) after 4h of plating, and all conductive tracks of all test pieces completed conduction after 6h of plating.

6.4 Bending Complexity in Three-Dimensional Space

We used four spatially conductive lines to test the method's capability in complex spatial scenarios (Figure 5c). Except for the first test block, which underwent two spatial transitions, the remaining went through three spatial transitions, whose corresponding conducting trace lengths are shown in (Figure 5c). Their resistance was measured post-plating. We made ten test pieces of each type, which were plated with a resistance of 0.5Ω or less.

7 APPLICATION

7.1 Complex Multi-Layer Internal Circuits

The primary advantage of our method is assisting users in integrating complex internal three-dimensional conductive structures. In

this example (Figure 6a), all controllers and light-emitting diodes are located at various spatial positions. Users can control the LEDs in four different directions by touching the four buttons at the top. Our auxiliary tools facilitate the easy completion of conductive lines, cavities, and other accessory modeling, allowing users to determine and create the direction of the lines efficiently.

7.2 High-Power Handheld Devices

Traditionally, the conductive capacity limitations in the personal manufacturing field of interactive objects have restricted their use to mainly sensing and display applications. However, owing to the current carrying capacity of the copper plating layer, our method supports high-voltage, high-power applications. This capability ensures that internal lines are protected from external environmental interference during operation, enabling the use of high-voltage handheld devices safely. In this case, the voltage is 30V, and the current is 0.59A (Figure 6b).

7.3 Enhancing the Functionality of Objects

Our method enables the rapid enhancement of an object's electrical functionality. For instance, by embedding high conductivity coils inside an ordinary cell phone holder, we transformed it into an integrated wireless charging base (Figure 6c). This example demonstrates how our approach can effectively add advanced electrical capabilities to everyday items.

8 CONCLUSIONS AND FUTURE WORK

Our work introduces a novel, cost-effective, integrated fabrication method for creating highly conductive interactable objects. This method employs multi-material printing to construct internal conductive trajectories with low initial conductivity, which are then enhanced through selective plating. This process culminates in the completion of the interactable objects. Notably, HCI researchers do not require expensive, specialized equipment for fabricating objects with complex internal conductive structures. Instead, a readily accessible consumer-grade multi-color 3D printer and basic plating equipment suffice. Additionally, a series of tests have been conducted to investigate our method's capabilities and provide design guidelines for internal conductive structures. We have also designed and produced three applications to demonstrate our method's effectiveness.

Looking ahead, we aim to enhance the functionality of our design tools. This will involve adding features such as wiring connection assistance, component libraries, and circuit design tips, increasing usability, and broadening our user base. We also explore more suitable and accessible materials and circuit placement methods to improve circuit resolution. Currently, the longest distance we have achieved with plating is 150mm. Larger interactive objects and longer plated circuits may necessitate extended plating times, which warrants further investigation.

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